Lecture Notes

Lecture 6

Diagonalization, Cayley–Hamilton, Functions of Operators

Lecture 5 found the eigenvalues and eigenvectors of an operator and showed that, with enough independent eigenvectors, its matrix becomes diagonal. This lecture turns that observation into a working theory. We characterize exactly when an operator is *diagonalizable*, prove the *Cayley–Hamilton theorem* (every operator satisfies its own characteristic equation), and use diagonalization to compute *functions of operators*—above all the exponential e^{tA}, which solves linear differential equations and, in the form $e^{-i\hat{H}t/\hbar}$, generates time evolution in quantum mechanics.

Throughout this lecture V is finite-dimensional over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and every matrix represents an endomorphism of such a space, unless an extension is explicitly labelled otherwise. Characteristic polynomials and matrix power series below belong to this finite setting; Lecture 14 develops bounded and unbounded Hilbert-space functional calculus.

The unifying idea is simple: in an eigenbasis an operator is just a list of independent scalings, so anything we can do to numbers—raise to a power, take an exponential, apply a function—we can do to the operator one eigenvalue at a time.

Any ket used in the quantum previews below is only Dirac notation for a vector: $|v\rangle$ means v, and the matrix columns are the images of ordered basis vectors. Bras and the inner-product rules are introduced in Lecture 7.

Learning objectives

By the end of this lecture you will be able to:

- Decide diagonalizability and diagonalize a matrix when possible.
- Distinguish algebraic from geometric multiplicity in the diagonalizability criterion.
- Compute functions of a diagonalizable operator—powers, e^{tA}, $f(T)$.
- State and apply Cayley–Hamilton and the spectral mapping rule.

Minimal-polynomial criteria are retained only for *extension recognition*; they are not a core use, practice, or progress requirement.

1 Diagonalizable operators

Definition 6.1 (Diagonalizable). An operator $T \in \mathcal{L}(V)$ is *diagonalizable* if V has a basis consisting of eigenvectors of T; equivalently, if $\mathcal{M}(T)$ is a diagonal matrix in some basis.

The two formulations agree because the matrix of T in a basis $v_1, \dots, v_n$ is diagonal exactly when $Tv_k = \lambda_k v_k$ for each k, i.e. when every basis vector is an eigenvector. The following theorem collects the practical tests.

Theorem 6.2 (Conditions for diagonalizability). Let $T \in \mathcal{L}(V)$ with $\dim V = n$ and distinct eigenvalues $\lambda_1, \dots, \lambda_m$. The following are equivalent.

1. T is diagonalizable.
2. V has a basis of eigenvectors of T.

> 3. $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$.
> 4. $\dim V = \dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T)$.

Proof. (1)⇔(2) is the definition. For (2)⇒(4): a basis of eigenvectors splits into vectors lying in the various eigenspaces, so $\sum_k \dim E(\lambda_k, T) \geq n$; the reverse inequality always holds because the eigenspaces form a direct sum (Lecture 5), giving equality. For (4)⇒(3): the sum $E(\lambda_1, T) + \cdots + E(\lambda_m, T)$ is direct (Lecture 5), so its dimension is $\sum_k \dim E(\lambda_k, T) = n = \dim V$; a subspace of full dimension is all of V. For (3)⇒(2): concatenating bases of the eigenspaces yields a basis of V consisting of eigenvectors. ∎

> **Corollary 6.3 (Distinct eigenvalues suffice).** If $T \in \mathcal{L}(V)$ has $\dim V$ distinct eigenvalues, then T is diagonalizable.

Proof. The corresponding eigenvectors are independent (Lecture 5), and an independent list of length $\dim V$ is a basis; condition (2) holds. ∎

Distinctness is sufficient but not necessary: the identity has the single eigenvalue 1 yet is already diagonal. What fails for a *defective* operator is condition (4)—some eigenspace is too small.

Example 6.4 (A diagonalizable operator). $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ has distinct eigenvalues $3, 1$ (Lecture 5), so by Corollary 6.3 it is diagonalizable, with eigenvectors $(1, 1), (1, -1)$ forming a basis. ◁

Example 6.5 (A non-diagonalizable operator). The shear $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ has only the eigenvalue 1, with $E(1, T) = \operatorname{span}\big((1, 0)\big)$ one-dimensional. The sum of eigenspace dimensions is $1 < 2$, so condition (4) fails and T is not diagonalizable: there is no basis of eigenvectors. ◁

Example 6.6 (A repeated eigenvalue that still diagonalizes). The matrix $A = \left(\begin{smallmatrix} 3 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{smallmatrix}\right)$ has eigenvalues $3, 2, 2$. The eigenspace $E(2, A)$ solves $(A - 2I)v = 0$, i.e. $v_1 + v_2 = 0$, which is *two*-dimensional: $E(2, A) = \operatorname{span}\big((-1, 1, 0), (0, 0, 1)\big)$. Together with $E(3, A) = \operatorname{span}\big((1, 0, 0)\big)$ the eigenspace dimensions sum to $1 + 2 = 3$, so A is diagonalizable despite the repeated eigenvalue. With $P = \left(\begin{smallmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$,

$$P^{-1}AP = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} = \operatorname{diag}(3, 2, 2).$$

Contrast the shear of Example 6.5, where the repeated eigenvalue had only a one-dimensional eigenspace. ◁

Example 6.7 (A complete diagonalization). Diagonalize $A = \left(\begin{smallmatrix} 4 & 1 \\ 2 & 3 \end{smallmatrix}\right)$. The characteristic polynomial $\lambda^2 - 7\lambda + 10 = (\lambda - 2)(\lambda - 5)$ gives eigenvalues 5 and 2; solving $(A - 5I)v = 0$ yields the eigenvector $(1, 1)$ and $(A - 2I)v = 0$ yields $(1, -2)$. Assembling these as the columns of P,

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}, \quad P^{-1} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}, \quad P^{-1}AP = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}.$$

The recipe—characteristic polynomial, eigenvectors, assemble P—works for every operator possessing a full set of eigenvectors. ◁

2 Diagonalization in practice

Let T on $\mathbb{F}^n$ be diagonalizable, with eigenvectors $v_1, \dots, v_n$ and eigenvalues $\lambda_1, \dots, \lambda_n$. Collect the data into

$$P = \begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix} \quad \text{(eigenvectors as columns)}, \qquad D = \operatorname{diag}(\lambda_1, \dots, \lambda_n).$$

Since $Av_k = \lambda_k v_k$ reads $AP = PD$ column by column, and P is invertible,

$$A = PDP^{-1}. \tag{6.1}$$

This is the change of basis of Lecture 4, specialized to an eigenbasis (Figure 1). Its great virtue is that powers and, soon, functions collapse to the diagonal:

$$A^k = (PDP^{-1})^k = PD^kP^{-1} = P\ \operatorname{diag}(\lambda_1^k, \dots, \lambda_n^k)\ P^{-1},$$

the middle factors $P^{-1}P$ telescoping away.

$$\begin{tikzcd} \mathbb{F}^n \arrow[r, "A"] & \mathbb{F}^n \\ \mathbb{F}^n \arrow[u, "P"] \arrow[r, "D = \operatorname{diag}(\lambda_k)"'] & \mathbb{F}^n \arrow[u, "P"'] \end{tikzcd}$$

Figure 1: In the eigenbasis the operator acts as the diagonal D; conjugating by P recovers $A = PDP^{-1}$. Powers and functions are applied to D entrywise.

Example 6.8 (Computing a power). For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ with $P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, $P^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, and $D = \operatorname{diag}(3, 1)$,

$$A^k = P\begin{pmatrix} 3^k & 0 \\ 0 & 1 \end{pmatrix}P^{-1} = \frac{1}{2}\begin{pmatrix} 3^k + 1 & 3^k - 1 \\ 3^k - 1 & 3^k + 1 \end{pmatrix}.$$

A single diagonalization gives every power at once; computing A^{10} directly would be far more work. $\triangleleft$

Example 6.9 (Fibonacci numbers by diagonalization). The recurrence $F_{n+1} = F_n + F_{n-1}$ is $\binom{F_{n+1}}{F_n} = A\binom{F_n}{F_{n-1}}$ with $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, so $\binom{F_{n+1}}{F_n} = A^n\binom{1}{0}$. The eigenvalues of A are the golden ratio $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$, and $A^n = PD^nP^{-1}$ yields Binet's closed form

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}.$$

Diagonalization turns a recurrence into a formula. $\triangleleft$

Example 6.10 (Long-run behavior of a Markov chain). The stochastic matrix $A = \begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix}$ has eigenvalues 1 and 0.5. Expanding any initial state in the eigenbasis, A^n leaves the $\lambda = 1$ component fixed and shrinks the $\lambda = 0.5$ component by $0.5^n \to 0$. Hence $A^n x_0$ converges to the $\lambda = 1$ eigenvector—the stationary distribution $(0.6, 0.4)$—for any initial probability distribution x_0. Diagonalization exposes the long-run behavior immediately. $\triangleleft$

Remark 6.11 (Decoupling). Physically, passing to the eigenbasis is *decoupling*. A system of coupled linear differential equations $\dot{x} = Ax$ becomes, in eigen-coordinates $y = P^{-1}x$, the uncoupled set $\dot{y}_k = \lambda_k y_k$. These first-order modes evolve independently: real parts produce growth or decay and imaginary parts produce oscillation. Vibrating *normal modes* belong to the second-order mechanical system in the next example.

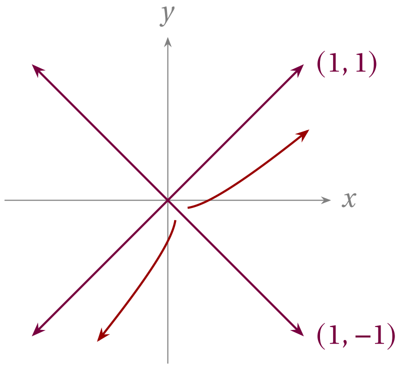

Figure 2: Phase portrait of $\dot{x} = Ax$ for $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ (eigenvalues 3, 1): the plotted curves are exact solutions with fixed signs of both eigen-coordinates. Generic forward orbits approach one of the two rays of the dominant eigendirection $(1, 1)$; the subdominant eigenline is exceptional.

Example 6.12 (Normal modes of two coupled masses). Two equal masses sit between fixed supports, with three identical springs: one from each support to its adjacent mass and one between the masses. In unit-mass, unit-spring coordinates they obey $\ddot{x} = -Kx$ with $K = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ in suitable units. The eigenvalues of K are 1 and 3, with eigenvectors $(1, 1)$ and $(1, -1)$. In these *normal coordinates* the system decouples into two independent oscillators: the in-phase mode $(1, 1)$ at angular frequency $\omega = 1$ and the out-of-phase mode $(1, -1)$ at $\omega = \sqrt{3}$. Diagonalizing K *is* finding the normal modes. ◁

3 The Cayley–Hamilton theorem

A striking fact ties an operator to its characteristic polynomial $\chi_T(\lambda) = \det(\lambda I - T)$.

Theorem 6.13 (Cayley–Hamilton). Every operator T on a finite-dimensional real or complex vector space satisfies its own characteristic equation: $\chi_T(T) = 0$.

Proof for diagonalizable T. Write $A = PDP^{-1}$ with $D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$. For any polynomial p, $p(A) = P\,p(D)\,P^{-1}$ and $p(D) = \operatorname{diag}(p(\lambda_1), \dots, p(\lambda_n))$. Taking $p = \chi_A$ and using that each λ_k is a root of χ_A,

$$\chi_A(A) = P\ \operatorname{diag}\big(\chi_A(\lambda_1), \dots, \chi_A(\lambda_n)\big)P^{-1} = P\ \operatorname{diag}(0, \dots, 0)\,P^{-1} = 0. \qquad \blacksquare$$

Proof in general. Over $\mathbb{C}$ every operator has an upper-triangular matrix (Lecture 5, where triangularizability is sketched) with diagonal entries $\lambda_1, \dots, \lambda_n$, and then $\chi_T(\lambda) = (\lambda - \lambda_1)\cdots(\lambda - \lambda_n)$. The triangular-product lemma of Lecture 5 gives $(T - \lambda_1 I)\cdots(T - \lambda_n I) = 0$ for such an operator, and that product is exactly $\chi_T(T)$, so $\chi_T(T) = 0$.

For a real operator, apply the proved complex result to its complexification. The matrix identity $\chi_T(T) = 0$ has real coefficients and the same real matrix entries, so it descends to the original real space. ■

Corollary 6.14 (Inverses and powers as low-degree polynomials). If T is invertible with $\chi_T(\lambda) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0$ (so $c_0 = (-1)^n \det T \neq 0$), then

$$T^{-1} = -\tfrac{1}{c_0}\big(T^{n-1} + c_{n-1}T^{n-2} + \cdots + c_1 I\big).$$

More generally every power T^k with $k \geq n$ reduces to a polynomial in T of degree $< n$.

Proof. Cayley–Hamilton gives $T^n + \cdots + c_1 T + c_0 I = 0$. Solve for $c_0 I$ and multiply by $T^{-1}/(-c_0)$ for the first claim; rearranging the same relation expresses T^n (hence inductively any higher power) through lower ones. ∎

Example 6.15 (Inverse from Cayley–Hamilton). For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, $\chi_A(\lambda) = \lambda^2 - 4\lambda + 3$, so $A^2 - 4A + 3I = 0$ and $A^{-1} = \frac{1}{3}(4I - A) = \frac{1}{3}\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$, which one checks satisfies $AA^{-1} = I$. ◃

Example 6.16 (Cayley–Hamilton, verified by hand). For $A = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right)$, $\chi_A(\lambda) = \lambda^2 - 5\lambda - 2$, and indeed

$$A^2 - 5A - 2I = \begin{pmatrix} 7 & 10 \\ 15 & 22 \end{pmatrix} - \begin{pmatrix} 5 & 10 \\ 15 & 20 \end{pmatrix} - \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

The operator annihilates its own characteristic polynomial. ◃

Example 6.17 (Reducing a high power). From $A^2 = 5A + 2I$ in Example 6.16, every higher power collapses to a combination of A and I: $A^3 = A \cdot A^2 = 5A^2 + 2A = 5(5A + 2I) + 2A = 27A + 10I$, and inductively $A^k = \alpha_k A + \beta_k I$ for scalars α_k, β_k. Cayley–Hamilton replaces unbounded powers with arithmetic in the two “coordinates” I and A. ◃

Remark 6.18 (Extension recognition: minimal polynomial). The monic polynomial of least degree annihilating T is its *minimal polynomial* m_T; it divides χ_T. Its roots that lie in the scalar field $\mathbb{F}$ are precisely the eigenvalues of T over $\mathbb{F}$; over $\mathbb{R}$, irreducible quadratic factors may remain. The field-correct criterion is

$$T \text{ is diagonalizable over } \mathbb{F} \iff \begin{array}{c} \scriptstyle m_T(x) \text{ splits over } \mathbb{F} \\ \scriptstyle \text{as a product of distinct linear factors.} \end{array}$$

For the real quarter-turn, $m_T(x) = x^2 + 1$ is square-free but does not split over $\mathbb{R}$, so the operator is not diagonalizable over $\mathbb{R}$; after complexification it splits as $(x - i)(x + i)$ and the operator diagonalizes over $\mathbb{C}$. By contrast, for the shear $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ the minimal polynomial is $(x - 1)^2$—since $A - I \neq 0$ but $(A - I)^2 = 0$—so its repeated factor obstructs diagonalizability, as in Example 6.5. Finally, if $\dim V = n$ and T has n distinct eigenvalues $\lambda_1, \ldots, \lambda_n$ in $\mathbb{F}$, then $m_T(x) = \prod_{k=1}^{n}(x - \lambda_k)$, confirming Corollary 6.3.

4 Functions of operators

We have already used *polynomials* of an operator. For a diagonalizable T on an $\mathbb{F}$-space we can apply a same-field function to the eigenvalues.

Definition 6.19 (Function of a diagonalizable operator). If $T = PDP^{-1}$ with $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ and $f : \operatorname{spec}(T) \to \mathbb{F}$, set

$$f(T) = P\, f(D)\, P^{-1}, \qquad f(D) = \operatorname{diag}\big(f(\lambda_1), \ldots, f(\lambda_n)\big).$$

This agrees with the power-series definition whenever the series converges, and is independent of the choice of eigenbasis.

A complex-valued function of a real operator instead acts after complexification: extend T to $T_{\mathbb{C}} \in \mathcal{L}_{\mathbb{C}}(V_{\mathbb{C}})$ and obtain $f(T_{\mathbb{C}}) \in \mathcal{L}_{\mathbb{C}}(V_{\mathbb{C}})$, not an endomorphism of the original real space in general.

The most important example is the *exponential*, defined for every finite matrix by the norm-convergent series

$$e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!}.$$

For diagonalizable $A = PDP^{-1}$ this sums to $e^A = P\ \mathrm{diag}(e^{\lambda_1}, \dots, e^{\lambda_n})\, P^{-1}$. Its defining property is the one that matters for dynamics.

> **Proposition 6.20 (Exponential and linear flow).** For fixed A, the function $t \mapsto e^{tA}$ satisfies $\dfrac{d}{dt} e^{tA} = A\, e^{tA}$ and $e^{0A} = I$. Hence $x(t) = e^{tA} x_0$ is the unique solution of $\dot{x} = Ax$ with $x(0) = x_0$.

Proof. Differentiate the series term by term (it converges uniformly on bounded t-intervals): $\frac{d}{dt} \sum_k \frac{t^k A^k}{k!} = \sum_{k \geq 1} \frac{t^{k-1} A^k}{(k-1)!} = A\, e^{tA}$. Then $\frac{d}{dt}\left(e^{tA} x_0\right) = A e^{tA} x_0$, and $x(0) = x_0$. This termwise step is licensed by the usual scalar exponential majorant in any matrix norm, uniformly on bounded t-intervals. For uniqueness, let $x(t)$ be any solution with the same initial value. Since A commutes with its power series,

$$\frac{d}{dt}\left(e^{-tA} x(t)\right) = -A e^{-tA} x(t) + e^{-tA} A x(t) = 0.$$

Thus $e^{-tA} x(t) = x_0$, and multiplication by e^{tA} gives $x(t) = e^{tA} x_0$. ■

Example 6.21 (Exponential of a 2×2 via the eigenbasis). With $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right) = PDP^{-1}$, $D = \mathrm{diag}(3, 1)$ as before,

$$e^{tA} = P \begin{pmatrix} e^{3t} & 0 \\ 0 & e^t \end{pmatrix} P^{-1} = \frac{1}{2} \begin{pmatrix} e^{3t} + e^t & e^{3t} - e^t \\ e^{3t} - e^t & e^{3t} + e^t \end{pmatrix}.$$

The solution of $\dot{x} = Ax$ is a combination of the modes e^{3t} and e^t along the eigendirections $(1, 1)$ and $(1, -1)$. ◁

Example 6.22 (Solving an initial value problem). For the same A with $x(0) = (1, 0)$, expand the initial data in the eigenbasis, $x(0) = \frac{1}{2}(1, 1) + \frac{1}{2}(1, -1)$, evolve each mode by its own exponential, and recombine:

$$x(t) = \tfrac{1}{2} e^{3t}(1, 1) + \tfrac{1}{2} e^t (1, -1) = \tfrac{1}{2}\left(e^{3t} + e^t,\ e^{3t} - e^t\right).$$

The dominant mode $e^{3t}(1, 1)$ dictates the long-time behavior, and the answer agrees with the first column of e^{tA} found above. ◁

Example 6.23 (Spin rotation: $e^{i\theta\sigma_x}$). Because $\sigma_x^2 = I$, the exponential series splits into even and odd powers:

$$e^{i\theta\sigma_x} = \sum_j \frac{(i\theta)^{2j}}{(2j)!} I + \sum_j \frac{(i\theta)^{2j+1}}{(2j+1)!} \sigma_x = \cos\theta\, I + i \sin\theta\, \sigma_x.$$

The conventional spin-$\frac{1}{2}$ rotation through physical angle ϕ is $R_x(\phi) = \exp(-i\phi\sigma_x/2)$. Hence the displayed expression corresponds to $\phi = -2\theta$: both the sign and the factor of two are part of the physical convention. The same algebraic calculation with any operator squaring to I produces a cos/sin pair. ◁

Example 6.24 (A square root of an operator). Since $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right) = PDP^{-1}$ with $D = \mathrm{diag}(3, 1)$ and positive eigenvalues, it has a square root

$$\sqrt{A} = P\ \mathrm{diag}(\sqrt{3}, 1)\, P^{-1} = \frac{1}{2} \begin{pmatrix} \sqrt{3} + 1 & \sqrt{3} - 1 \\ \sqrt{3} - 1 & \sqrt{3} + 1 \end{pmatrix},$$

which satisfies $(\sqrt{A})^2 = A$. A function of an operator is no harder than the same function of its eigenvalues. ◁

Remark 6.25 (Identities transport to operators). Any scalar identity among functions holds for diagonalizable operators, since both sides act eigenvalue by eigenvalue. The two displayed identities in fact hold for every finite matrix directly: multiply the absolutely convergent power series for e^A and e^{-A} (all powers of A commute) to get I, and use the even/odd power-series definitions of sine and cosine to obtain $\sin^2 A + \cos^2 A = I$. No density claim is needed.

Proposition 6.26 (Spectral decomposition). If T is diagonalizable with distinct eigenvalues $\lambda_1, \dots, \lambda_m$, there are projections $P_1, \dots, P_m$ (each onto an eigenspace along the others) with

$$P_i P_j = 0 \ \ (i \neq j), \qquad P_1 + \cdots + P_m = I, \qquad T = \lambda_1 P_1 + \cdots + \lambda_m P_m,$$

and consequently $f(T) = \sum_k f(\lambda_k) P_k$ for any $f : \operatorname{spec}(T) \to \mathbb{F}$.

Proof. In the eigenbasis T is constant ($= \lambda_k$) on each eigenspace block; let P_k be the projection picking out the $E(\lambda_k, T)$ block. These satisfy the stated relations, $T = \sum_k \lambda_k P_k$, and applying f blockwise gives $f(T) = \sum_k f(\lambda_k) P_k$. ■

This is the finite-dimensional shadow of the spectral theorem of Lecture 10, in which the P_k become *orthogonal* projections.

Example 6.27 (Spectral decomposition in coordinates). For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ the eigenprojections onto $E(3) = \operatorname{span}(1,1)$ and $E(1) = \operatorname{span}(1,-1)$ are

$$P_1 = \tfrac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad P_2 = \tfrac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix},$$

with $P_1 + P_2 = I$ and $P_1 P_2 = 0$. Then $A = 3P_1 + P_2$, and Proposition 6.26 gives $A^k = 3^k P_1 + P_2$ and $e^A = e^3 P_1 + e P_2$ with no matrix multiplication at all—recovering Example 6.8 instantly. ◁

5 Spectral mapping and time evolution

Functions act on the spectrum in the obvious way.

Proposition 6.28 (Spectral mapping). If T is diagonalizable over $\mathbb{F}$ and $f : \operatorname{spec}(T) \to \mathbb{F}$, then $\operatorname{spec}\big(f(T)\big) = \{f(\lambda) : \lambda \in \operatorname{spec}(T)\}$, and every eigenvector of T for λ is an eigenvector of $f(T)$ for $f(\lambda)$.

Proof. If $Tv = \lambda v$ then $f(T)v = Pf(D)P^{-1}v$ acts on the eigenvector v by the scalar $f(\lambda)$, since in eigen-coordinates v is a standard basis vector and $f(D)$ scales it by $f(\lambda)$. Conversely, the same eigenbasis diagonalizes $f(T)$, and its diagonal entries are exactly the values $f(\lambda_k)$. Therefore no additional eigenvalues of $f(T)$ occur. ■

The converse inclusion for eigenvectors fails when f is not injective on the spectrum: for $T = \operatorname{diag}(1,-1)$ and $f(\lambda) = \lambda^2$ we get $f(T) = I$, every nonzero vector of which is an eigenvector, while T has only two eigendirections. Distinct eigenvalues can be merged by f, and merging enlarges an eigenspace.

Example 6.29 (Spectrum of a function). For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ with spectrum $\{1, 3\}$, the operator e^A has spectrum $\{e, e^3\}$ and A^{10} has spectrum $\{1, 3^{10}\}$, all with the same eigenvectors $(1,-1)$ and $(1,1)$. Once the eigenvectors are known, no function of A requires recomputing them. ◁

Example 6.30 (Oscillation from imaginary eigenvalues). The generator $A = \left(\begin{smallmatrix} 0 & -\omega \\ \omega & 0 \end{smallmatrix}\right)$ has eigenvalues $\pm i\omega$, and a short computation (or the cos/sin split, since $A^2 = -\omega^2 I$) gives

$$e^{tA} = \begin{pmatrix} \cos \omega t & -\sin \omega t \\ \sin \omega t & \cos \omega t \end{pmatrix}.$$

So $\dot{x} = Ax$ is rotation at angular frequency ω: imaginary eigenvalues produce oscillation, exactly as real ones produced the growth and decay of Example 6.21. This is the linear-algebra heart of simple harmonic motion. ◁

Remark 6.31 (Quantum time evolution). In the finite-dimensional, time-independent, closed-system model, a state vector evolves by $|\psi(t)\rangle = e^{-i\hat{H}t/\hbar}|\psi(0)\rangle$. In an energy eigenbasis $\hat{H} = \operatorname{diag}(E_1, \dots, E_n)$, so

$$e^{-i\hat{H}t/\hbar} = \operatorname{diag}\big(e^{-iE_1 t/\hbar}, \dots, e^{-iE_n t/\hbar}\big) :$$

each stationary state merely acquires a phase $e^{-iE_n t/\hbar}$, and a general state is a superposition of these independently rotating phases. Diagonalizing the Hamiltonian solves this finite pure-point dynamics. General spectral integrals and domain questions are deferred to Lecture 14.

Remark 6.32 (Finite/discrete simultaneous diagonalization). Commuting diagonalizable operators can be brought to diagonal form in a *single* basis—the common eigenvectors of Lecture 5. Physically, finite-dimensional commuting self-adjoint observables share an orthonormal eigenbasis, so a state can have simultaneously definite values for both. For general Hilbert-space observables the corresponding statement uses joint spectral measures and projection-valued measures rather than necessarily an ordinary eigenbasis; Lecture 14 gives that outlook.

6 Extension: when diagonalization fails

The entire section is optional extension material. Jordan form, Jordan–Chevalley decomposition, defective exponentials, exceptional points, and ladder/cohomology analogies do not count toward core practice or progress.

Not every operator is diagonalizable, but the generalized eigenvectors of Lecture 5 restore order. The obstruction is a *nilpotent* operator—one with $N^k = 0$ for some k—the part of an operator that no change of basis can diagonalize away. On a single defective eigenvalue $A = \lambda I + N$ with N nilpotent, and since λI and N commute the *nilpotent factor* e^{tN} terminates:

$$e^{tA} = e^{\lambda t}\, e^{tN} = e^{\lambda t}\Big(I + tN + \tfrac{t^2}{2!}N^2 + \cdots + \tfrac{t^{k-1}}{(k-1)!}N^{k-1}\Big).$$

For the defective shear block $A = \left(\begin{smallmatrix} \lambda & 1 \\ 0 & \lambda \end{smallmatrix}\right)$, the nilpotent part $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ satisfies $N^2 = 0$, so

$$e^{tA} = e^{\lambda t}\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}.$$

The extra factor t is the signature of a defective eigenvalue: it produces solutions $t\, e^{\lambda t}$, the same ones that appear for repeated roots of a linear differential equation.

Theorem 6.33 (Jordan normal form · cited). Every operator on a finite-dimensional complex space has a basis of generalized eigenvectors in which its matrix is block diagonal, each block a *Jordan block*

$$J = \lambda I + N, \qquad N = \begin{pmatrix} 0 & 1 & & \\ & 0 & \ddots & \\ & & \ddots & 1 \\ & & & 0 \end{pmatrix},$$

carrying the eigenvalue λ on the diagonal and the nilpotent shift N on the superdiagonal. Each block is one Jordan chain of generalized eigenvectors (Lecture 5).

The shift N sends each generalized eigenvector to the one below it, $e_j \mapsto e_{j-1} \mapsto \cdots \mapsto e_1 \mapsto 0$—a ladder that descends to zero in finitely many steps, which is exactly what nilpotency means.

Theorem 6.34 (Jordan–Chevalley decomposition • cited). On a complex vector space of finite dimension, every operator A splits uniquely as

$$A = D + N, \qquad DN = ND,$$

with D diagonalizable and N nilpotent, and both D and N polynomials in A.

Because D and N are polynomials in A, they commute with everything A commutes with, so the splitting respects any symmetry A has—this is what makes it more than a normal-form trick. It is also the cleanest route to $e^{tA} = e^{tD}e^{tN}$ for an arbitrary matrix: the diagonal factor supplies the oscillation and decay, the nilpotent factor the polynomial-in-t corrections.

Remark 6.35 (Nilpotents and ladder operators). Nilpotent operators are the ladder operators of physics, seen in finite dimensions. On a spin-j representation (dimension $2j+1$) the raising operator J_+ steps $|m\rangle \mapsto |m+1\rangle$ until it annihilates the top state, so $J_+^{2j+1} = 0$: J_+ is nilpotent, and its action is precisely the shift N of a Jordan block. The harmonic-oscillator lowering operator truncated to N levels is nilpotent in the same way ($a^N = 0$), and the oscillator ladder of Lectures 13–14 is its infinite-dimensional limit. Nilpotency squared to zero, $Q^2 = 0$, is also the defining relation of the supercharge in supersymmetric quantum mechanics, of the BRST charge, and of the exterior derivative—the algebra behind cohomology.

Remark 6.36 (Where defective operators are physical, and a note on self-study). In finite dimension, self-adjointness rules defectiveness out entirely (Lectures 9–10), so finite observables are diagonalizable. General self-adjoint operators instead use spectral measures and need not have an ordinary eigenbasis. Defective operators belong instead to *non-Hermitian* physics: the critically damped oscillator, whose repeated root gives the tell-tale $t\,e^{-\gamma t}$ above, and the *exceptional points* of PT-symmetric and open systems, where two eigenvectors coalesce and the Hamiltonian becomes a genuine Jordan block—a measurable effect in optics and cavity physics. For a *self-adjoint* family $H_0 + \varepsilon V$ no such coalescence can occur at real ε; it can at complex ε, and how near those points come to the real axis is exactly what limits the perturbation series of optional Lecture 15. The Jordan and Jordan–Chevalley results are offered here for self-study: we state their structure without proving the construction, and carry forward only the principle—diagonal where possible, a nilpotent correction where not.

Summary of main concepts

On a finite-dimensional real or complex space, an operator is diagonalizable iff it has a basis of eigenvectors, equivalently iff its eigenspace dimensions sum to $\dim V$; having $\dim V$

distinct eigenvalues is a convenient sufficient condition. In an eigenbasis $A = PDP^{-1}$, so powers and functions are computed one eigenvalue at a time: $A^k = PD^kP^{-1}$ and $f(T) = Pf(D)P^{-1}$ for $f : \mathrm{spec}(T) \to \mathbb{F}$, with the finite matrix exponential e^{tA} solving $\dot{x} = Ax$. Every such operator satisfies its own characteristic equation $\chi_T(T) = 0$ (Cayley–Hamilton), which expresses inverses and high powers as low-degree polynomials in T. Spectral mapping sends eigenvalues λ to $f(\lambda)$; in quantum mechanics this is exactly why diagonalizing the Hamiltonian solves the time evolution $e^{-i\hat{H}t/\hbar}$.

- **Diagonalizable**—has an eigenbasis $\iff$ eigenspace dimensions sum to $\dim V$; $\dim V$ distinct eigenvalues suffice.
- **Diagonalized form**—$A = PDP^{-1}$, so powers and functions act one eigenvalue at a time; e^{tA} solves $\dot{x} = Ax$.
- **Cayley–Hamilton**—$\chi_T(T) = 0$; expresses inverses and high powers as low-degree polynomials in T.
- **Spectral mapping**—f sends eigenvalue λ to $f(\lambda)$—why diagonalizing H solves $e^{-iHt/\hbar}$.
- **Jordan form**—extension recognition only: the canonical form when diagonalization fails (self-study; not core progress).

Exercises for this lecture are in the companion sheet `exercises-6.pdf`.